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Rescaled-range and power spectrum analyses on well-logging data

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SUMMARY

Fractal theory and related developments have significantly enlightened our understanding of many natural phenomena and led to an upsurge in research efforts in many different disciplines. This has unfortunately caused some ambiguities, misunderstandings, and even misuses in the published literature concerning some new concepts in fractal theory and their actual implementations. As new applications and ideas arise almost daily, it is important to present the methods in a common framework to minimize further confusion. In this paper, we scrutinize and compare two of the most frequently applied methods for characterizing fractal time-series, namely the rescaled-range (R/S) and the power spectrum method, and describe their proper applications in wire line well-logging data. We outline the problems related to applications of these two methods. Based on numerical analysis on fractional Gaussian noise (fGn) and fractional Brownian motion (fBm), we demonstrate the clear necessity to distinguish between fGn-like and fBm-like data series. For fBm-like data series, the use of their successive increments is proposed against using various other alternatives in R/S analysis in order to make the estimated Hurst exponent (Hu) comparable with the global scaling exponent (H) from power spectrum analysis. We argue that well-logging data are generally analogous to fBm, and thus we use their incremental series rather than raw well data themselves to estimate Hu when applying R/S. We also reveal a connection between the transient zone length in the R/S technique and the shortest-wavelength that can be included for a linear regression in the power spectrum method.

Key words: borehole geophysics, Fourier transforms, fractals, geostatistics, numerical techniques.

1 INTRODUCTION

Careful studies have revealed some problems in the current application of rescaled-range analysis (R/S) and the power spectrum in determining the Hurst exponent (Hu) and the fractal dimension (D). In particular, it appears to be evident that R/S and power spectrum methods rarely give similar estimates of Hu, and in many cases differ significantly if not performed with particular care. Individually, both R/S and the spectrum method have been criticized for their inaccuracies (Fox 1989; Hough 1989; Lo 1991; Bassingthwaighte & Raymond 1994; North & Halliwell 1994). However, discussions on the obvious discrepancies between the two are often safely ignored. It is our view that neglecting this issue could hamper true understanding of Hu and D, and could make less meaningful estimates. This paper, through a detailed mathematical analysis and case studies, is meant to clarify some of the ambiguities involved in their applications and make results from different authors and/or techniques more consistent and comparable.

The concept of the Hurst exponent was first introduced by Hurst (1951) via his well-known empirical relationship, called the rescaled-range technique, in studying long-term discharge variations of the River Nile, and later publicized and solidified by Mandelbrot & Wallis (1969) and Feder (1988). Since then it has gained popularity and been applied in many subjects, including network traffic, bioengineering, physical simulation and of course, geology and geophysics. The key point of this technique is the following empirical relationship:

$$R/S = (a\tau)^{Hu}, \tag{1}$$

where a is a constant, R is the difference between the maximum and minimum accumulated departure from the mean within time span τ, S is the standard deviation serving as a scaling factor and Hu is the Hurst exponent coined by Mandelbrot (1982). The accumulated departure with respect to the mathematical mean $\langle\xi\rangle$ of a time-series ξ_i is

$$X(t) = \sum_{i}^{N}(\xi_i - \langle\xi\rangle), \tag{2}$$

where t is time, i is the time sample index and N is the number of time samples. The range R is defined as

$$R(\tau) = \max_{1\le t\le\tau} X(t) - \min_{1\le t\le\tau} X(t) \tag{3}$$

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and $S(\tau)$ is the standard deviation

$$S(\tau) = \left[\frac{1}{\tau} \sum_{t=1}^{\tau} (\xi_t - \langle \xi \rangle)^2 \right]^{1/2}. \quad (4)$$

Eq. (1) clearly reveals a scaling-invariant property of the data with respect to the period τ. Specifically, the rescaled range R/S is linearly proportional to $a\tau$ on a log–log plot, the slope of which is equal to Hu.

Among various definitions for a fractal time-series, one of the commonly used ones is

$$F(\sigma t) = \sigma^H F(t). \quad (5)$$

It describes a self-affine process, i.e. a subset of a fractal process is a scaled version of the fractal set itself. We call H in eq. (5) the global scaling exponent and σ is a scaling operator. For a self-affine time-series defined by eq. (5), it can be shown that its power spectral density $S(f)$ has a power-law dependence on frequency f:

$$S(f) \propto f^{-\beta}, \quad (6)$$

where we call β the spectrum power. Turcotte (1997) showed that the following relationship between H and β exists:

$$\beta = 2H + 1. \quad (7)$$

We are now confronted with a dilemma as to whether the global scaling exponent (H) derived from eq. (7) is equal to the Hurst exponent (Hu) from the R/S technique. In many studied cases no distinctions are made between H and Hu, and they are often used interchangeably (Hewett 1986; Shiomi *et al.* 1997; Dolan *et al.* 1998; Taqqu & Teverovsky 1998), while in a few situations H is differentiated from Hu (Turcotte 1997). This conflict has posed confusion and misinterpretation in estimating Hu. It has been observed that estimating Hu and H for the same data series, but using two different techniques above, can give significantly different results. This prompts us to consider in what sense H and Hu can be considered equivalent.

2 IS H EQUAL TO Hu?

To answer this question it is helpful to discuss two basic fractal processes that are closely related: the fractional Brownian motion (fBm) and fractional Gaussian noise (fGn). fGn is the successive steps or increments of fBm; in other words, the accumulative summation of fGn gives fBm. While fBm is non-Gaussian and non-stationary, fGn is Gaussian and stationary.

It is straightforward to prove that for fBm with spectrum power $1 \le \beta_{\mathrm{fBm}} \le 3$, the corresponding fGn has $-1 \le \beta_{\mathrm{fGn}} \le 1$. An fBm series $X(t)$ is nowhere differentiable, but in a distribution when the subsequent interval $\Delta X = X(t_i) - X(t_{i-1})$ is sufficiently small compared with the total length of X, from the definition of the derivative we can write

$$G(t) = \Delta X = X(t_i) - X(t_{i-1}) \cong \Delta t X'(t), \quad (8)$$

where Δt is a time sampling interval. Taking the first derivative of $X(t)$ in the time domain is equivalent to multiplying $X(f)$, which is the Fourier transform of $X(t)$, by $-i2\pi\ f$ (Bracewell 1965), $i = (-1)^{1/2}$. By taking the Fourier transform of both sides of (8), we have

$$G(f) \cong \Delta t[-2\pi i f X(f)]. \quad (9)$$

Here we see that the non-differentiability of fBm can actually be relaxed in the frequency domain. The power spectra are given by

$$|G(f)|^2 \cong |\Delta t[-i2\pi f X(f)]|^2 = -4\Delta t^2 \pi^2 f^2 |X(f)|^2. \quad (10)$$

Here $|\ G(f)|^2$ is the power spectrum of the fGn and $|X(f)|^2$ is the power spectrum of the fBm. Because $|X(f)|^2 \propto f^{-\beta}$, we have $|G(f)|^2 \propto f^{-\beta+2} = f^{-(\beta-2)}$. This proves that

$$\beta_{\mathrm{fGn}} \cong \beta_{\mathrm{fBm}} - 2. \quad (11)$$

Or from eq. (7),

$$H_{\mathrm{fGn}} \cong H_{\mathrm{fBm}} - 1. \quad (12)$$

The proof of eqs (11) and (12) reveals that β and H are indicators of the roughness of the data. The larger H is the smoother the data. By taking a first-order derivative we increase the roughness of the data but decrease the value of H by 1. Now eq. (7) can be extended to:

$$\begin{aligned} \beta_{\mathrm{fBm}} &= 2H_{\mathrm{fBm}} + 1 \\ \beta_{\mathrm{fGn}} &= 2H_{\mathrm{fGn}} + 1 \cong 2(H_{\mathrm{fBm}} - 1) + 1 = 2H_{\mathrm{fBm}} - 1. \end{aligned} \quad (13)$$

While H varies from 1 to -1 from fBm to fGn, we realize from eqs (1)–(4) that Hu will always lie between 0 and 1 (Mandelbrot & Van Ness 1968; Mandelbrot & Wallis 1969). Thus, in general, H should not be considered as equivalent to Hu. Ignoring the difference between H and Hu can lead to confusing results. What has often been assumed is to make $Hu_{\mathrm{fBm}} = Hu_{\mathrm{fGn}} = H_{\mathrm{fBm}}$, and because of this, eq. (3) is often unjustifiably simplified to

$$\begin{aligned} \beta_{\mathrm{fBm}} &= 2Hu + 1 \\ \beta_{\mathrm{fGn}} &= 2Hu - 1. \end{aligned} \quad (14)$$

In reality, $Hu_{\mathrm{fGn}} \approx H_{\mathrm{fBm}}$ but $Hu_{\mathrm{fBm}} \neq Hu_{\mathrm{fGn}}$ and $Hu_{\mathrm{fBm}} \neq H_{\mathrm{fBm}}$. Eq. (14) is to keep Hu unchanged from $1 \le \beta \le 3$ to $-1 \le \beta \le 1$ and is actually a sacrifice, realizing that the meaning of Hu has been altered from $1 \le \beta \le 3$ to $-1 \le \beta \le 1$, because theoretically the relation $\beta = 2H + 1$ holds to be true for $-\infty < \beta < +\infty$. In order for eq. (14) to be meaningful, one has to assume that Hu of fBm is Hu from fGn, the successive increments of fBm. Without this common consensus, results of R/S analysis from different authors are incompatible and significant differences exist between H and Hu. To avoid these problems we have to clearly identify whether the time-series under study is analogous to fBm or fGn. If it is analogous to a fBm with spectrum power $1 \le \beta \le 3$, we have to find the subsequent increments of this time-series and then apply R/S to the new incremental series. Otherwise, as will be shown later in the next section, H will not equal Hu and Hu is almost certainly larger than 0.85, making Hu a very limited quantity to be used in characterizing fBm-like time-series. This is equivalent to saying that, in a strict sense, Hu is only better suited for capturing the properties of noise, such as fGn with Hu between 0 and 1, but not for random walks such as fBm.

Because of the delicate relationship between Hu and H, extra caution needs to be taken in relating Hu and H to the fractal dimension (D). For self-affine time-series, Voss (1985) derived the following approximate relationship between the spectrum power β and the fractal dimension D for fBm ($1 \le \beta \le 3$):

$$\beta = 5 - 2D. \quad (15)$$

From eq. (7), we have the much celebrated relationship

$$D = 2 - H. \quad (16)$$

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A direct analytical relationship between D and Hu has not been observed. However, for fGn, because $Hu_{\rm fGn} \approx H_{\rm fBm}$, eq. (16) can be extended to Hu so we can have

$$D \approx 2 - Hu. \tag{17}$$

This relationship cannot be extended for fBm because $Hu_{\rm fBm} \neq H_{\rm fBm}$. In order for eq. (17) to be applicable for fBm, we first need to obtain the associated fGn from fBm, calculate Hu from fGn, and take it as Hu of fBm.

We wish to point out that practically there is nothing wrong with applying R/S to fBm directly, but the Hu so obtained is not comparable to H from the power spectrum. In geophysical applications such as reservoir simulations, we value H as a better quantity than Hu because H captures the complexities of the data more accurately, while Hu is confined between 0 and 1. It is more realistic to say that R/S approximates the true scaling exponent H in eq. (5), but care must be taken when we relate H to Hu because these two quantities are not always interchangeable. The R/S technique can be used as a secondary method to roughly estimate H, mainly to validate the results from the spectrum method.

3 NUMERICAL SIMULATIONS

Using a frequency domain algorithm, we simulate 21 fBm series, each with 4096 data points. The input H for fBm ranges from 0 to 1 with a constant interval of 0.05. We could also simulate 21 fGn series using the same algorithm, but for the purpose of this article, we obtain the 21 fGn series directly from the 21 fBm series by taking successive increments. We then apply the R/S method to both fGn and fBm series to estimate Hu. It is expected that the H value from the power spectrum should be very close to the input H, for the simulation is itself based on a spectrum algorithm. It is nevertheless meaningful for us to check the robustness of our power spectrum method by estimating H on simulated fBm with the spectrum method. We also apply a third independent method called the variance of increment method to estimate H and validate our simulation.

Table 1 shows the results of analysis on simulated data. A rescaled-range analysis is applied on both fGn and fBm. We can immediately realize that R/S analysis on fBm almost surely gives an estimate of $Hu > 0.85$, as predicted. On the other hand R/S-estimated Hu from fGn is close to the estimate from the power spectrum and the variance of the increment analysis on fBm (Fig. 1). This further proves our earlier statement that, in order to make the estimate of Hu from R/S comparable to H from the power spectrum, we should apply R/S to the corresponding fGn series of fBm rather than to fBm itself. The final two columns of Table 1 show the spectrum power β of fBm and fGn. We can see that the relationships between β and Hu for fBm, $\beta_{\rm fBm} \approx 2Hu + 1$, and for fGn, $\beta_{\rm fGn} \approx 2Hu - 1$ are approximately true for our simulations, when the correct meaning of Hu is assigned.

Fig. 1 shows clearly that the rescaled range tends to overestimate the lower H values (for $H \lesssim 0.7$) and underestimate high H values (for $H \gtrsim 0.7$). This agrees well with results from other studies (Mandelbrot & Wallis 1969; Bassingthwaighte & Raymond 1994). Fig. 1 also shows an obvious linear correlation between Hu and H. An empirical relationship between Hu and H for fBm/fGn based on the least-squaresx regression is

$$H = 1.23Hu - 0.16. \tag{18}$$

Eq. (18) provides a simple way to roughly estimate the true scaling exponent H from a rescaled-range analysis for fGn/fBm. It is now straightforward to determine the crossover point where $H = Hu$. We find that when $H = Hu \approx 0.7$, the two methods should be able to give nearly identical results.

The initial 'transient zone' for the R/S technique persists when the input H is less than or equal to 0.5 (Fig. 2). Visually, the length of the transient zone is increasing from a lag spacing of 31 data points to approximately 63 data points when H increases from 0 to 0.5. The gradient of this zone is always higher than that of the rest of the data.

We find that when input H becomes larger than 0.5, the transient zone can no longer be clearly discernible from the rest, and it is safe to include this zone to calculate Hu. Meanwhile, however, another

Table 1. Results from R/S and power spectrum analyses on simulated fBm/fGn.

fBm no	Input H	R/S on fGn (Hu)	R/S on fBm (Hu)	Power spectrum (H)	Variance of increment (H)	β of fBm	β of fGn
1	0.0	0.173	0.900	0.012	0.054	1.024	−0.783
2	0.05	0.155	0.890	0.043	0.034	1.087	−0.763
3	0.1	0.226	0.938	0.114	0.139	1.228	−0.572
4	0.15	0.261	0.952	0.186	0.186	1.372	−0.452
5	0.2	0.309	0.947	0.206	0.219	1.411	−0.397
6	0.25	0.285	0.939	0.248	0.164	1.497	−0.327
7	0.3	0.323	0.970	0.324	0.272	1.648	−0.171
8	0.35	0.365	0.965	0.370	0.373	1.740	−0.104
9	0.4	0.443	0.995	0.411	0.371	1.822	0.000
10	0.45	0.508	0.996	0.463	0.529	1.927	0.117
11	0.5	0.569	1.004	0.504	0.531	2.009	0.197
12	0.55	0.580	0.991	0.541	0.561	2.081	0.261
13	0.6	0.676	0.999	0.631	0.576	2.261	0.407
14	0.65	0.670	0.993	0.654	0.575	2.308	0.485
15	0.7	0.764	0.999	0.700	0.756	2.399	0.579
16	0.75	0.786	0.997	0.772	0.702	2.543	0.714
17	0.8	0.784	1.000	0.807	0.688	2.614	0.763
18	0.85	0.789	0.966	0.849	0.637	2.699	0.855
19	0.9	0.837	1.005	0.922	0.714	2.844	1.003
20	0.95	0.863	1.000	0.986	0.618	2.971	1.097
21	1.0	0.886	1.000	1.012	0.542	3.024	1.188

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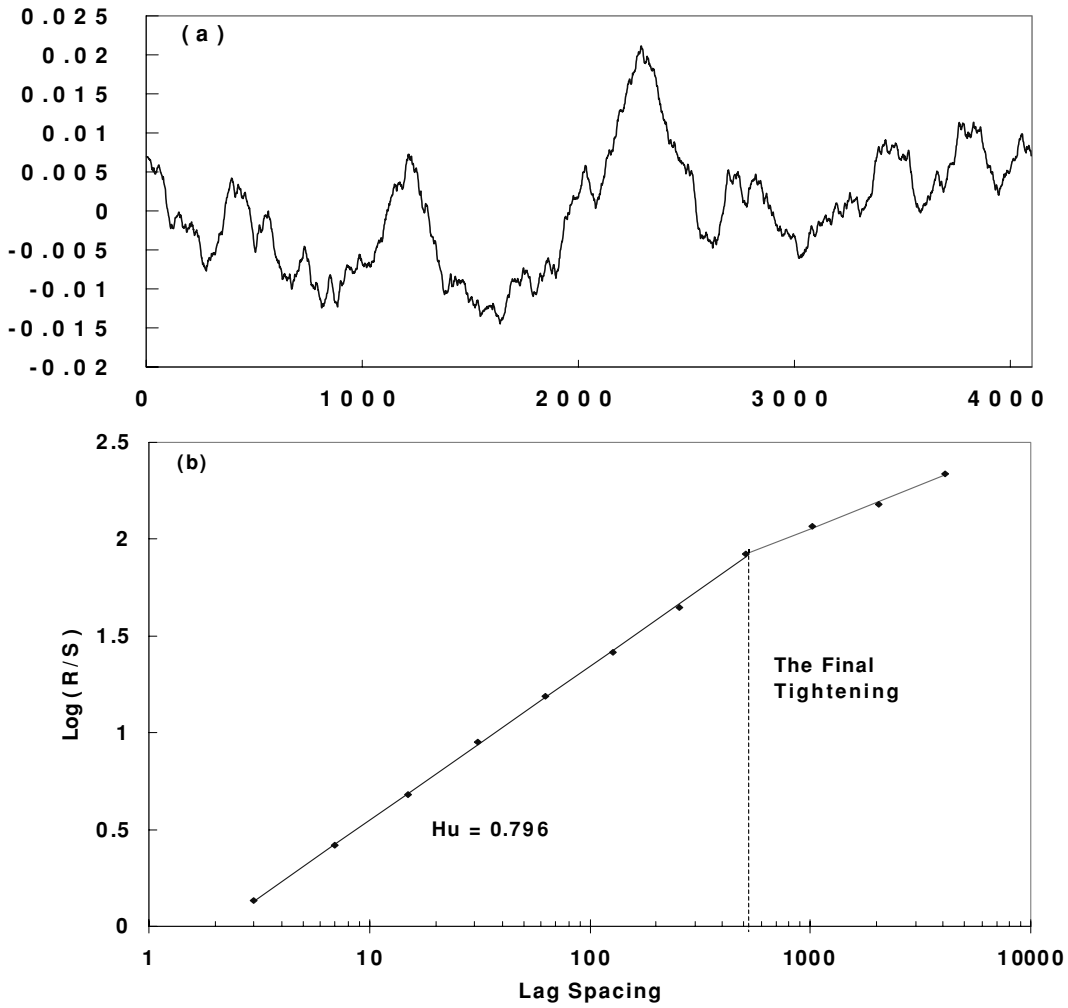


Figure 3. Plots showing the final tightening in the *R*/*S* analysis. (a) Simulated fBm no 18 with input $H = 0.85$. (b) *R*/*S* analysis on the simulation.

4.1 Gamma-ray data

The data, sampled at a constant rate of 0.5 ft for a total interval of 1984 ft, is from a well located in Panoma gas field, Grant County, southwest Kansas. Formation rocks penetrated by this well include shale, sandstone, siltstone, gypsum, anhydrite and dolomite. Fig. 4 shows the raw gamma-ray data (a), normalized data (b), data after moving average trend removal (c) and successive increments (d) from the raw data. To the right of these data series in Fig. 4 are their respective histograms. The raw data does not have a normal distribution, but all the others in the panel have distributions close to normal distributions. Unlike the raw gamma-ray data, we observe that modified data sets can be considered to be Gaussian/normal and stationary. Note here that normalization is not simply to extract the mean from the data and then divide the data by its standard deviation, because this will have little effect on removing short-wavelength variability. In fact, we estimate the time-varying variance and find the trend of the variance of non-stationary time-series data. We then normalize the data using the estimated variance.

Fig. 5 shows the rescaled range analysis on all data sets in Fig. 4. First, we see no transient zone for raw gamma-ray data, but distinct transient zones of various lengths (n) exist for modified data sets (b, c and d). Incremental series (d) gives the shortest transient zone. All observable transient zones are overlapping with the *R*/*S* plot of the raw data; in other words, transient zones from different data sets (b, c and d) have the same gradient on the *R*/*S* plot. For the raw gamma-ray data (Fig. 6a), the blindness of *R*/*S* to the transient zone may have an adverse effect on the *Hu* estimation if *R*/*S* is applied directly to raw data. This may also help to explain why various modifications to the raw data are necessary. Secondly, *R*/*S* analyses on modified data sets (b, c and d) give similar estimates of *Hu* regardless of how the raw data are modified. In this example, $Hu_{\text{increment}}$, $Hu_{\text{normalized}}$ and Hu_{MA} are 0.518, 0.378 and 0.54, respectively. On the other hand, *Hu* estimated from raw data is 0.91, which is substantially higher. Finally, we find that the incremental data (d) gives the best estimate of *Hu* because it has the shortest transient zone and the highest correlation coefficient ($R = 0.998$) from least-squares regression. In contrast, data with normalization (b) gives the lowest correlation coefficient of 0.955 and data with moving average trend removal (c) has the fewest data points for regression and has a correlation coefficient of 0.993. This leads to our conclusion that, while all the modified data give close estimates of *Hu*, the incremental series from the raw data gives the best estimate of *Hu*.

Unlike simulated fGn/fBm, well data suffer from the moving average effect of the logging tools (Shiomi *et al.* 1997). This effect will induce an abrupt drop of power at large wavenumbers (or frequencies) on the power spectrum plot (Fig. 6). The location of this breaking point is dependent on the resolution of the logging tools.

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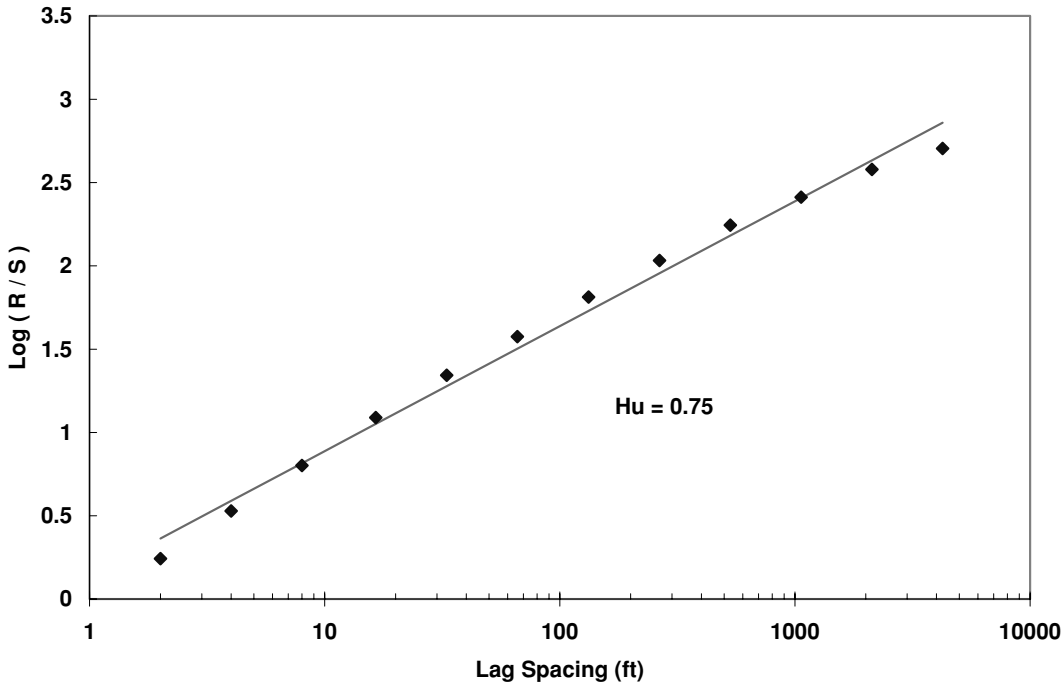


Figure 10. An example showing that, although a good linear regression ($R = 0.99$) can be made without identifying the existence of a transient zone, the estimated *Hu* is too high compared with H.

Table 2. *Hu*, H and D calculated from two well-logging data. (D is the fractal dimension.)

		Raw	Normalized	MA	Increment	Spectrum (H)
GR	*Hu*	0.91	0.38	0.54	0.52	0.27
	$D = 2 - Hu$ (or H)	1.09	1.62	1.46	1.48	1.73
	Box counting D_b	1.37				
	Regularization D_r	1.65				
	Transient zone lengths (ft)	0	70	94.6	9.3	
Sonic	*Hu*	0.89	0.51	0.3	0.25	0.017
	$D = 2 - Hu$ (or H)	1.11	1.49	1.7	1.75	1.983
	Box counting D_b	1.60				
	Regularization D_r	1.94				
	Transient zone lengths (ft)	0	212.7	170.3	11.2	

between $D_b = 1.60$ and $D_r = 1.94$. However, using $D = 2 - Hu$ based on the raw data, the estimated D is only 1.11, much smaller than D_b and D_r. D_b and D_r are also reasonably close to the fractal dimension D (=1.983) from $D = 2 - H$. This once again demonstrates that R/S analysis directly on the raw data is not appropriate for determining the fractal dimension.

5 CONCLUSIONS AND SUGGESTIONS

In this paper, we examined the paradoxical results often found in the literature concerning the Hurst exponent estimation, and provided an acceptable framework for more accurate analyses. Our arguments were made based on theoretical considerations, analyses from simulations of a group of fGn/fBm and studies on real data examples. We gained a better understanding of the nature of the well log data and argued that raw well-logging data should be considered as fBm-type time-series, which have a spectrum power $\beta \geq 1$. Consequently, for fBm-like data, in order to make the results from rescaled-range and power spectrum techniques consistent and comparable, and to make meaningful fractal dimension estimates, we need to use their incremental series, rather than the raw time-series themselves for R/S analysis. R/S analyses applied directly on the raw well data always tend to give erroneously high estimates of *Hu* (>0.85), making the technique itself less useful. On the other hand, R/S analyses on the incremental series give estimates of *Hu* close to H from power spectrum analyses on the raw data.

Other types of data modification, such as normalization or trend removal, could also be used to improve the analysis, and these modified new data series including the incremental series can give similar estimates of *Hu*. However, the increments have the shortest transient zones that make R/S analyses more accurate. It is also more objective to use the incremental series, because for each raw data series there exists only one unique incremental series that is also very easy to obtain. Data from normalization or trend removal could vary a lot depending on the specific data processing procedures. The degree of accuracy can also be viewed on the power spectrum plot where the increments have the broadest wavenumber/frequency ranges for linear regression.

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Since these modifications to raw well data introduce transient zones in the *R*/*S* analysis, it is very important to identify this zone and exclude it from calculating *Hu*; otherwise it will also result in erroneously high estimates of *Hu*. Comparing with the power spectrum analysis, we find that the transient zone lengths are closely connected to the regression lengths at long-wavelength region in the power spectrum. This reveals the close tie between these two independent data analysis methods.

Fractal models have been used for reservoir simulation (Hewett 1986; Hewett & Behrens 1990; Goggin *et al.* 1992). The decision in choosing the correct Hu/H is very important. For example, a realization based on $Hu = 0.11$ (Holliger 1996) will be quite different from a realization using $Hu = 0.7$ (Leary 1991), even for the same sonic data in Cajon Pass, CA. While we argue that raw well-logging data are of fBm type, in practice, for a new unknown data series it is wise to use the power spectrum method first to determine whether the spectrum power β is larger or smaller than 1. If $\beta \geq 1$, the data need to be considered as fBm-like and we use its increments for *R*/*S* analysis. There might be cases where the calculated $\beta < 1$ even for fBm-type data. This is most probably caused by an insufficient number of data points in the calculation. Both *R*/*S* and spectrum methods require a sufficient number of registered points for an accurate estimation. At least two independent methods are necessary for an accurate scaling analysis because they can provide a double-checking mechanism.

Finally, we realize that the Hurst exponent (*Hu*) should not be confused with the global scaling exponent (*H*) in practice. While these two quantities have some connections, they are fundamentally different and can only be considered equivalent under some constraints. The many discrepancies and conflicts in reported studies are a result of a lack of genuine understanding of the nature of both the data and the analysing techniques.

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